

SyllabAI: A Knowledge-Augmented RAG System Design for Syllabus-Grounded Exam Preparation

An AI-Augmented Learning Management System for IGCSE/IAL in the Bangladesh English-Medium Context

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Abstract

The official mark schemes for every Edexcel and Cambridge IGCSE/IAL paper are freely available online. The syllabi are public. The examiner reports are published. Every piece of information needed for diagnostic, personalised exam preparation exists, yet in Bangladesh’s English-medium sector, where coaching centres have hollowed out the senior sections of schools and where private candidates study without any institutional support, the infrastructure to use this information at scale does not. Students drill past papers reactively, fail, retake with the same strategy, and fail again.

This paper presents the design of SyllabAI, an AI-augmented learning management system that converts the existing public assessment infrastructure into personalised, diagnostic instruction. The system integrates four components: a syllabus knowledge graph that maps prerequisite relationships and surfaces foundation gaps; a student proficiency overlay combining Bayesian knowledge tracing with Ebbinghaus forgetting-curve decay; a retrieval-augmented generation (RAG) chatbot that grounds its explanations in official mark schemes using chain-of-thought reasoning and a Text2Cypher knowledge-graph query step; and a learning log whose schema is designed to detect, at scale, each of the six struggle types identified in a companion paper (Khondokar, 2026). The architecture, which I call Knowledge-Augmented RAG (KA-RAG), combines the structural reasoning of the knowledge graph with the grounded generative capabilities of the RAG chatbot, ensuring that every response is both pedagogically appropriate and traceable to an official source. The paper describes the system architecture, the design decisions motivated by each struggle type, and pre-registered predictions about telemetry signatures and the mediated effect of system use on examination outcomes. The system has not yet been deployed; this is a design proposal and pre-registration, not an empirical study.

Keywords: Retrieval-Augmented Generation, Knowledge Graph, Knowledge-Augmented RAG, Bayesian Knowledge Tracing, Intelligent Tutoring Systems, Learning Management System, IGCSE, International A-Level, Bangladesh, Design-Based Research

1 Introduction

The International General Certificate of Secondary Education (IGCSE) and International Advanced Level (IAL) qualifications offered by Pearson Edexcel and Cambridge Assessment International Education are the primary academic pathway for English-medium students in Bangladesh seeking international university admission. Despite the rapid growth of this educational segment, the infrastructure for diagnostic, personalised exam preparation is largely absent. Private tutors and coaching centres operate on a reactive model: students are drilled on past papers in the weeks preceding an examination, with little diagnostic attention to prerequisite knowledge gaps or to the cognitive processes that examination questions are designed to elicit. Where a student fails a paper, the standard response is more drilling, not diagnostic remediation.

This pattern is particularly striking because the IGCSE and IAL syllabi are public, well-structured documents, and because official mark schemes, examiner reports, and grade boundaries are all freely available. The information required for diagnostic teaching exists; what is missing is the infrastructure to use it at scale and at the level of individual students. Three specific gaps motivate the system design proposed in this paper.

First, the *multi-topic tagging gap*: many Bangladesh students use chapter-wise past-paper compilations that sort questions by single topic, but these compilations silently drop multi-topic questions (questions that draw on content from multiple syllabus topics simultaneously) because such questions cannot be cleanly assigned to a single chapter. Students who rely exclusively on these compilations are therefore unprepared for the cross-topic questions that distinguish A from A* grades.

Second, the *mock-exam deficit*: private candidates, who constitute a substantial fraction of the Bangladesh English-medium population, do not get school-organised mock exams. Content knowledge without exam-condition practice is fragile: a student who knows the content perfectly but has never practised under timed conditions will, on a hard paper, leave marks unattempted that they could have scored.

Third, the *K.E.A.R.N. Gap* (The Brilliant Brains, 2025): exams do not measure what students know but what students can demonstrate in writing under time pressure using mark-scheme-specific language. This “exam literacy” is a hidden curriculum that schools, particularly in developing-country contexts, do not teach explicitly. The official mark schemes published by Edexcel and Cambridge are, in effect, detailed formative assessment rubrics (Black and Wiliam, 1998; Hattie and Timperley, 2007), yet in the typical coaching-centre context they are used summatively (to score) rather than formatively (to diagnose and remediate).

1.1 Research Question

This paper is organised around a single research question:

RQ. To what extent can an AI-augmented learning management system, integrating a syllabus knowledge graph, a Bayesian knowledge tracing proficiency overlay, and a knowledge-augmented retrieval-augmented generation (KA-RAG) chatbot grounded in official mark schemes, serve as both an intervention that addresses diagnosed student struggles and a research instrument that detects those struggles at scale?

1.2 Companion Paper and Diagnostic Target

This system design assumes a typology of student struggles developed in a companion paper (Khondokar, 2026). That paper proposes a four-layer model of academic performance (Content Knowledge $\times$ Exam Literacy $\times$ Learning Strategy $\times$ Self-Regulation) and a six-type typology of struggles: Type 1 (prerequisite gap), Type 2 (surface engagement), Type 3 (exam literacy), Type 4 (metacognitive), Type 5 (motivational), and Type 6 (instructional-environment). Each struggle type has a hypothesised behavioural signature observable through interviews, school records, or learning-platform telemetry. The present paper takes that typology as its diagnostic target: the system is designed to detect each struggle type at scale and to provide targeted intervention for each.

The four-layer model and the typology are summarised briefly here for readability; for the full theoretical derivation, see Khondokar (2026). The four layers are: Layer 1 (Content Knowledge, what the student actually knows), Layer 2 (Exam Literacy, the ability to convert knowledge into marks), Layer 3 (Learning Science, the use of effective study strategies and metacognitive monitoring), and Layer 4 (Self-Regulation, the capacity to maintain effort and persist through difficulty). The six struggle types map onto the layers: Type 1 is a Layer 1 failure (prerequisite gaps), Type 2 is a Layer 3 failure (surface-level engagement), Type 3 is a Layer 2 failure (exam literacy), Type 4 is a Layer 3/4 failure (metacognitive), and Type 5 is a Layer 4 failure (motivational and mindset). The system described in this paper is designed to address Layers 2, 3, and 4 directly, and to surface Layer 1 gaps for teacher-mediated remediation.

1.3 Significance

This paper makes three contributions. First, it presents the design of a KA-RAG architecture specifically tailored to high-stakes examination preparation, combining a syllabus knowledge graph with retrieval-augmented generation in a way that has not, to my knowledge, been previously proposed for the IGCSE/IAL context. Second, it specifies a learning-log schema whose fields are designed to detect each of the six struggle types at scale, providing a research instrument for the companion paper’s typology. Third, it pre-registers system-specific predictions about telemetry signatures and the mediated effect of system use on examination outcomes.

The remainder of this paper is structured as follows. Section 2 reviews the relevant literature on knowledge graphs, retrieval-augmented generation, and intelligent tutoring systems. Section 3 details the SyllabAI architecture. Section 4 describes the design-based research methodology for evaluating the system. Section 5 pre-registers system-specific predictions. Section 7 concludes.

2 Literature Review

2.1 Knowledge Graphs and Personalised Learning

Knowledge graphs have emerged as a powerful tool for representing and reasoning about the structural relationships within educational content. Tong and Ren (2025) presented a novel approach for personalised learning path generation by integrating deep knowledge trac-

ing with cognitive load estimation, demonstrating that knowledge-graph-based systems can effectively model prerequisite relationships between concepts and adapt learning pathways accordingly. Narayan et al. (2025) presented an AI-driven framework for personalised learning and remedial recommendation grounded in knowledge graphs, showing that such systems can identify specific knowledge gaps and recommend targeted remedial content with high precision.

Abdelmagied et al. (2025) introduced Educational Knowledge Graphs (EduKGs) as effective tools for extracting structured knowledge concepts from learning materials and linking them to assessment items. This work demonstrates the feasibility of constructing domain-specific knowledge graphs from syllabus documents and past examination papers, which is the foundation upon which SyllabAI’s knowledge graph is designed. Li et al. (2025a) provided a unified analytical framework for personalised learning path recommendation based on knowledge graphs, identifying knowledge state modelling, path planning, and adaptive adjustment as the core components of effective knowledge-graph-based learning systems.

2.2 Retrieval-Augmented Generation and LLMs in Education

Retrieval-augmented generation has emerged as a particularly promising architecture for educational applications because it addresses the hallucination problem that has limited the adoption of LLM-based chatbots in educational settings. Li et al. (2025b) analysed RAG across various educational scenarios, concluding that the architecture’s ability to ground generated responses in verified source material makes it uniquely suited for contexts where factual accuracy is critical, such as examination preparation. Swacha and Gracel (2025) specifically examined RAG chatbots for education, finding that the combination of retrieval and generation phases enables these systems to provide explanations that are both contextually relevant and citation-traceable, a capability that is essential for mark-scheme-aligned tutoring in the IGCSE/IAL context.

Létourneau et al. (2025) provided a systematic review of AI-driven intelligent tutoring systems (ITS) in K-12 education, finding that these systems show promising effects on student learning and performance, though the quality of experimental designs varies considerably. The review noted that the most effective ITS implementations are those that provide not only domain-specific content delivery but also adaptive feedback mechanisms that respond to individual student needs. The combination of LLMs with RAG has clear applications in education: a chatbot grounded in official past papers and mark schemes can answer student questions about specific examination items, explain why the mark scheme awards marks in a particular way, and walk students through the reasoning chain that connects a question to its answer.

Wei et al. (2022) showed that chain-of-thought prompting, in which the model is instructed to articulate intermediate reasoning steps, substantially improves performance on multi-step problems. The combination of RAG grounding with chain-of-thought prompting produces responses that are both verifiable, because each claim is traceable to a retrieved source, and pedagogically appropriate, because the reasoning is made visible to the student. A second technical development is the convergence of knowledge graphs with LLMs. Hu et al. (2024) surveyed graph retrieval-augmented generation, in which a structured knowledge graph is used alongside a vector store to retrieve both relevant text and relevant relational context. The combination, sometimes called Knowledge-Augmented RAG (KA-RAG), is the architecture adopted by SyllabAI. The foundational RAG and dense retrieval work (Karpukhin et al., 2020; Lewis et al.,

2020) and the large language model scaling work (Brown et al., 2020) provide the technical substrate on which KA-RAG builds.

2.3 Intelligent Tutoring and Knowledge Tracing

The technical literature on intelligent tutoring systems provides the machinery for operationalising the formative assessment and mastery learning traditions. Corbett and Anderson (1995) introduced Bayesian knowledge tracing (BKT), a hidden Markov model that maintains, for each skill and each student, a probability that the student has mastered the skill. Each observed response updates the probability, accounting for the possibilities of guessing, correct response despite no mastery, and slipping, incorrect response despite mastery. Koedinger et al. (1997) demonstrated that cognitive tutors based on BKT could produce substantial learning gains in real classroom settings, with effect sizes comparable to those of one-on-one human tutoring. Pane et al. (2015) reported evidence from a large-scale personalised learning study that systems incorporating such features produce measurable improvements in mathematics and reading achievement, although the heterogeneity of the interventions limits causal inference.

Positioning relative to ASSISTments, Cognitive Tutors, and Khan Academy. SyllabAI’s proficiency overlay is conceptually descended from the BKT machinery of ASSISTments (Feng et al., 2009) and the Cognitive Tutors (Anderson et al., 1995; Koedinger et al., 1997), and shares with Khan Academy’s mastery-learning system (Murphy et al., 2014) the goal of proficiency-anchored progression through a structured skill graph. The differences are substantive, however. ASSISTments and Cognitive Tutors are general-purpose tutoring platforms whose skill graphs are derived from domain-specific cognitive task analyses (typically US K–12 mathematics); SyllabAI’s skill graph is constructed from official Edexcel and Cambridge syllabi, with prerequisite edges validated against the specific mark-scheme demands of those examinations. Khan Academy’s mastery system uses a three-correct-in-a-row heuristic for proficiency and does not maintain a misconception-specific overlay; SyllabAI uses BKT with monthly recalibration and extends it to a Bayesian Diagnosis Tracing (BDT) model that tracks misconception-specific probabilities (see §3.6). None of the three prior systems grounds its explanations in retrieval-augmented generation over official mark schemes, which is the distinctive contribution of SyllabAI’s KA-RAG chatbot (see §3.4). None of the three maintains a learning-log schema designed to detect a multi-type struggle typology at scale, which is the distinctive contribution of SyllabAI’s learning log (see §3.5). The contribution of SyllabAI is therefore not the proficiency overlay in isolation (which is well-established in the prior literature) but the integration of a mark-scheme-grounded KA-RAG chatbot, a misconception-aware BDT overlay, a multi-topic-tagged question bank, and a struggle-typology-targeted learning log into a single system designed for the specific demands of IGCSE/IAL examination preparation.

2.4 Synthesis

The literature reviewed above suggests that the technical machinery for diagnosing and remediating weak sub-skills at scale exists, in the form of BKT-style proficiency tracking, prerequisite-aware knowledge graphs, and RAG chatbots grounded in official assessment materials. What has not been previously proposed, to my knowledge, is the integration of all three into a single system specifically designed for IGCSE/IAL examination preparation, with a

learning-log schema designed to detect a multi-type struggle typology at scale. The companion paper (Khondokar, 2026) provides the diagnostic target; the present paper provides the system that detects and addresses it.

3 The SyllabAI System Design

3.1 Architecture Overview

The SyllabAI system comprises four principal components: a syllabus knowledge graph, a student proficiency overlay, a RAG chatbot, and a learning log. Each component is described below, with particular attention to the design decisions that bear on the system’s dual role as intervention and research instrument. The system exposes two parallel interfaces over the same data layer: a student-facing interface for independent or teacher-managed study, and a teacher-facing interface for class management, diagnostic analytics, and professional development support (see §3.11). Figure 1 shows the four components and their data flows.

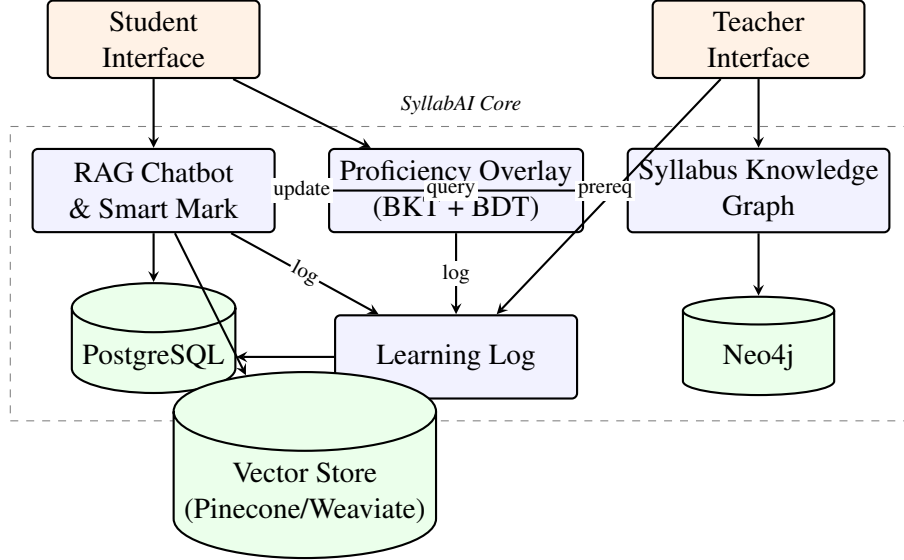


Figure 1: SyllabAI architecture: four principal components (RAG Chatbot & Smart Mark, Proficiency Overlay, Syllabus Knowledge Graph, Learning Log) and their data flows. Solid arrows show data flow; interfaces (student/teacher) sit above the core; storage (Neo4j, PostgreSQL, vector store) sits below.

3.2 The Syllabus Knowledge Graph

The syllabus knowledge graph is a structured representation of the official Edexcel and Cambridge syllabi. The ontology comprises four node types, Subject, Unit, Topic, Sub-topic, and three edge types, PART_OF, REQUIRES_PREREQUISITE, RELATED_TO. The closest precedent in the literature is EducOnto/EduKG (Hubert et al., 2022), which models university curricula with prerequisite relationships; SyllabAI’s KG differs by using exam-board syllabi (not university courses) and by adding a proficiency overlay that tracks individual student mastery per node. The graph is constructed by parsing official syllabus PDFs to extract the hi-

erarchical structure, then using a large language model to suggest prerequisite relationships, which are validated by subject matter experts before being committed to the graph. This hybrid automated-plus-human approach trades construction speed for accuracy and is consistent with the recommendations of Hu et al. (2024) and Abdelmagied et al. (2025) for educational knowledge graph construction. The structured, hierarchical presentation of the graph directly reduces extraneous cognitive load (Sweller, 1988; Sweller et al., 2011): by presenting topics in their prerequisite order and making relationships between concepts explicit, the graph eliminates the disorganised, fragmented presentation that characterises coaching-centre materials and that consumes working memory capacity needed for germane processing. Each sub-topic node carries metadata including the official syllabus code, the learning objectives, links to relevant past paper questions, via the question bank, and links to approved revision resources. Sub-topic nodes also carry child nodes representing common misconceptions identified in the examiner reports and the conceptual change literature (Posner et al., 1982): for example, the “Fractions” node has child misconceptions such as “adding denominators” and “forgetting common denominators”. Each wrong-answer distractor in the question bank is tagged with the specific misconception it indicates, enabling the system to distinguish a missing schema from a wrong schema (see §3.6 below). The graph is stored in Neo4j, which supports efficient traversal of prerequisite relationships and is well-suited to the visualisation needs of the student dashboard. Figure 2 summarises the ontology.

Syllabus Knowledge-Graph Schema

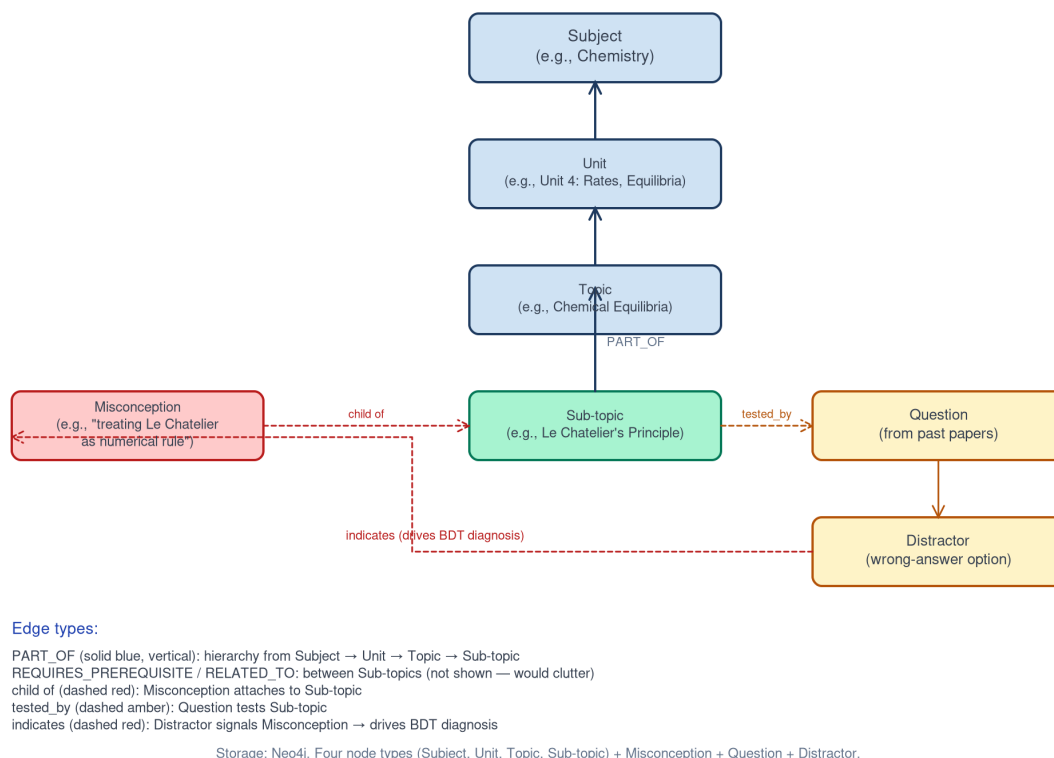


Figure 2: Syllabus knowledge-graph schema. Four node types (Subject, Unit, Topic, Sub-topic) form the PART_OF hierarchy. Sub-topic nodes carry child Misconception nodes (identified from examiner reports and the conceptual-change literature). Questions are tagged to the Sub-topics they test, and each Distractor is tagged with the specific Misconception it indicates — this distractor-to-misconception mapping is what drives the BDT diagnosis (Figure 4).

A recurring problem observed in my tutoring practice motivates a specific feature of the teacher-side knowledge graph visualisation. Guardians typically hire private tutors only in the last few months before the exam. By that point, the student has accumulated months or years of unaddressed knowledge gaps, and the tutor's first task is diagnostic: figure out what the student does and does not know. This takes time, often several sessions, and the student frequently does not know their own weak points, which makes the diagnosis harder. The SyllabAI teacher-side knowledge graph visualisation is designed to solve this problem directly. When a teacher opens a student's profile, they see, immediately, a colour-coded map of the student's proficiency at the sub-topic level. Red nodes are weak points. Amber nodes are borderline. Green nodes are mastered. The teacher can then plan targeted remediation from the first session, rather than spending the first month on diagnosis. In my experience, this single feature would save roughly 30 to 50 percent of the time currently wasted in early tutoring sessions. The teacher-side visualisation could also integrate with the existing teacher support resources provided by the examination boards: Cambridge International offers professional development materials and subject communities (Cambridge International Education, 2021), and the system could surface links to these resources for topics where the teacher's own knowledge is thin, directing them to the official training rather than leaving them to search independently.

A second design feature, also motivated by a concrete observation, is the multi-topic tagging of questions. Many Bangladesh students use chapter-wise question-paper compilations produced by Unique Coaching. These compilations sort past-paper questions by topic, which is genuinely useful for targeted revision. But there is a blind spot. Some exam questions draw on content from multiple topics simultaneously. A single Chemistry question might require knowledge of equilibrium, kinetics, and stoichiometry. Unique Coaching’s chapter-wise compilations skip these multi-topic questions, because they cannot be cleanly assigned to a single chapter. They include only single-topic questions. Students who rely exclusively on these compilations therefore never practise multi-topic questions. They are well-prepared for the within-topic portion of the exam and unprepared for the cross-topic portion. On exam day, the cross-topic questions are the ones that distinguish A from A*, and these students lose marks they did not know they would lose. SyllabAI also has topic-tagged questions, we parsed questions during the RAG ingestion pipeline and added topic metadata that powers features like Target Test and Test Builder, but we added secondary topic tags so that a question can be retrieved by any of the topics it tests, not just its primary one. The system can also generate a “cross-topic” practice set specifically containing multi-topic questions, which Unique Coaching’s compilations cannot do.

3.3 The Student Proficiency Overlay

For each student and each sub-topic node, the system maintains a proficiency score in the range $[0, 1]$. The score is updated through a Bayesian Knowledge Tracing (BKT) model (Corbett and Anderson, 1995), with time-based decay applied according to an Ebbinghaus forgetting curve. The spacing-effect literature (Cepeda et al., 2006) establishes that distributed practice produces substantially better retention than massed practice, and the forgetting-curve decay operationalises this directly: proficiency erodes predictably without active retrieval, and the system schedules review at the point of predicted decay, converting the spacing effect from a laboratory finding into a system feature.

Why BKT alone, not a hybrid. An earlier draft of this paper proposed a hybrid update pipeline combining rule-based updates (+0.05 correct, -0.03 incorrect, +0.02 streak bonus), a weighted moving average (0.7 previous + 0.3 latest), and nightly BKT batch updates. On reflection and reviewer feedback, the hybrid has been replaced by BKT alone for the persisted proficiency score, for three reasons. First, BKT is a coherent Bayesian belief-update model whose statistical properties (convergence, bias, variance) are well-characterised; combining it with rule-based and WMA updates produces a score whose joint statistical properties are not analysed in the literature and would require a separate simulation study to characterise. Second, BKT’s parameters (slip, guess, initial knowledge, learning rate) have clear interpretations, which is essential for the teacher-side XAI layer (Section 3.8): a teacher can understand “this student has slip = 0.15 on Fractions” but cannot easily understand a proficiency score that mixes three update schemes. Third, BKT alone is sufficient for the real-time UI feedback that the rule-based updates were intended to provide: the UI displays the prior BKT probability and the posterior after the latest response, which gives the student immediate feedback without polluting the persisted score.

Real-time UI feedback. Real-time UI feedback (the colour change of a knowledge-graph node immediately after a student answers a question) is computed by applying the BKT update synchronously with the response, using the current slip/guess/learning-rate parameters for that student–sub-topic pair. The persisted proficiency score (used for adaptive assessment routing, struggle-type detection, and teacher analytics) is computed by the same BKT update applied in the nightly batch job, which uses population-level slip/guess parameters calibrated on the accumulating data. The synchronous and batch updates use the same model, so the persisted score converges to the synchronous score over time; there is no discrepancy between what the student sees and what the system records, beyond the parameter-refinement lag.

BKT parameters, calibration, and forgetting-curve decay. BKT parameters are calibrated as follows. Initial knowledge (L_0) is set to 0.1 for all sub-topics at student intake (a low prior that is quickly overridden by observed responses). Slip and guess parameters are initialised to population-level defaults (slip = 0.1, guess = 0.25) and are recalibrated monthly via maximum-likelihood estimation on the accumulating response data, stratified by sub-topic. Learning rate (T) is initialised to 0.1 and is similarly recalibrated monthly. The forgetting curve is applied nightly to all sub-topics using the Ebbinghaus-exponential form $P(t) = P_0 \cdot e^{-t/\tau}$, where τ is the decay constant; τ is set to 30 days for sub-topics with current proficiency < 0.4 (faster decay for weaker knowledge), 90 days for proficiency $\in [0.4, 0.8)$, and 365 days for proficiency ≥ 0.8 (slower decay for stronger knowledge). These thresholds are consistent with the spacing-effect literature’s recommendation that review be scheduled at the point of predicted decay (Cepeda et al., 2006). While deep knowledge tracing (DKT) approaches using recurrent neural networks have shown higher predictive performance on large datasets (Piech et al., 2015), the system chose BKT for its interpretability and its suitability for small-sample contexts, which is the constraint of the first pilot cycle.

Simulation-based validation. Before deployment, the BKT-alone model is validated against the original hybrid model on synthetic data with known ground truth. A simulated student population ($N = 1,000$) is generated with known mastery trajectories on a 50-node knowledge graph; responses are sampled from a known generative model with slip = 0.1, guess = 0.25, learning rate = 0.1. Both the BKT-alone and the original hybrid models are fitted to the simulated responses, and the recovered proficiency trajectories are compared to the ground truth. Acceptance criteria: BKT-alone must achieve bias < 0.05 and RMSE < 0.10 on the recovered proficiency; if BKT-alone meets these criteria and the hybrid does not improve RMSE by more than 0.02, the simplification to BKT-alone is justified. The simulation code is deposited alongside the OSF pre-registration. The BKT model is further extended to a Bayesian Diagnosis Tracing (BDT) model that tracks misconception-specific probabilities rather than a single mastery probability; see §3.6 for details.

The proficiency overlay is visualised as a colour-coded knowledge graph on the student dashboard: green for proficiency above 0.8, amber for 0.4 to 0.8, red for below 0.4, and grey for unattempted. Clicking a node opens a panel showing the student’s recent attempts on that sub-topic, common mistakes, and recommended resources. The visualisation serves both motivational and diagnostic functions: it makes progress tangible and it surfaces weaknesses that the student might not otherwise notice. The prerequisite-aware design directly addresses Type 1 (prerequisite gap) struggles: when a student is red on “Integration”, the system traces the prerequisite chain and surfaces whether the root cause is a gap in “Differentiation”, “Alge-

braic Manipulation”, or “Fractions”. A richer diagnostic dimension is planned (Cycle 2) using the SOLO Taxonomy (Biggs and Collis, 1982), which describes five levels of understanding: prestructural (no understanding), unistructural (one relevant aspect), multistructural (several aspects but not connected), relational (aspects connected into a coherent structure), and extended abstract (generalised to new contexts). Instead of a single 0–1 proficiency score, each sub-topic node could carry a SOLO level, telling the system not just whether the student knows the topic but what kind of understanding they have. This is more diagnostically useful than a single number because it distinguishes different kinds of “partially learned” and prescribes different interventions.

3.4 The RAG Chatbot and Smart Mark

The RAG chatbot, internally referred to as the Past Paper Tutor, is a retrieval-augmented generation system grounded in official past papers, mark schemes, examiner reports, and approved revision notes. The architecture follows the standard RAG pipeline (Karpukhin et al., 2020; Lewis et al., 2020; Li et al., 2025b; Swacha and Gracel, 2025) with two modifications specific to the educational context.

The first modification is the use of a Text2Cypher step to extract entities from the student’s question and to query the syllabus knowledge graph before performing vector retrieval. When a student asks, for example, to explain question 4c on a specific past paper, the system extracts the paper identifier, the question identifier, and the relevant sub-topic, queries the knowledge graph to confirm the sub-topic and its prerequisites, and then performs vector retrieval over the chunked past paper corpus. The retrieved chunks, together with the graph context, are assembled into a prompt for the LLM, which is instructed to answer using only the provided context and to cite sources. This KA-RAG architecture, combining knowledge-graph structural retrieval with vector semantic retrieval, follows the approach surveyed by Hu et al. (2024) and demonstrated for educational contexts by Abdelmagied et al. (2025). *Note on terminology:* “KA-RAG” (Knowledge-Augmented RAG) is the label used in this paper for the architecture that Hu et al. (2024) surveys under the names “GraphRAG” and “GRAG” and that Edge et al. (2024) implements as the open-source GraphRAG pipeline. The different label is used here to emphasise the knowledge-graph structural component over the more general graph-RAG family and to maintain terminological continuity with the companion paper’s framework; no claim of architectural novelty beyond the surveyed prior work is intended. (Cycle 1 uses Text2Cypher; Cycle 2 will replace this with the GraphRAG pipeline (Edge et al., 2024), which enables multi-hop reasoning over the prerequisite chain and community-level summaries for the teacher dashboard. The transition requires re-indexing the document corpus with the GraphRAG pipeline, a one-time effort with increased LLM usage during indexing.)

Language input: English-only in Cycle 1, code-switching in Cycle 2. The Cycle 1 chatbot accepts English input only. This is a known equity limitation: a substantial fraction of Bangladesh English-medium students have Bangla as their first language and routinely code-switch between Bangla and English when discussing technical concepts with peers and tutors. A student who would naturally ask “eta ki kore korbo?” (“how do I do this?”) must reformulate the question in English, which may depress chatbot usage among Bangla-dominant students. Code-switching tolerance (Bangla–English mixed input) is planned for Cycle 2, to be implemented via a multilingual LLM (e.g., GPT-4o or a Llama-3 multilingual variant) with a

code-switching-aware system prompt. This is a Cycle 2 rather than Cycle 3 objective because the equity cost of the English-only limitation is recognised as material; the Cycle 1 system will report chatbot-usage rates stratified by student first-language to quantify the equity gap that Cycle 2 must close.

Text2Cypher accuracy and fallback (Cycle 1). Text2Cypher is an active research area with substantial accuracy challenges; the Cycle 1 system validates its accuracy on a held-out set of 200 student questions hand-labelled with ground-truth Cypher queries. Acceptance criteria: precision ≥ 0.85 and recall ≥ 0.80 on entity extraction (paper identifier, question identifier, sub-topic). When Text2Cypher fails (confidence below threshold or query returns empty), the system falls back to vector-only retrieval (skipping the knowledge-graph query), and the fallback is logged for later analysis. The fallback rate is reported in the Cycle 1 results paper; a fallback rate above 20% triggers a Cycle 2 redesign of the Text2Cypher step (potentially via fine-tuning on the accumulating labelled data, or via replacement with GraphRAG). Figure 3 summarises the full KA-RAG request pipeline.

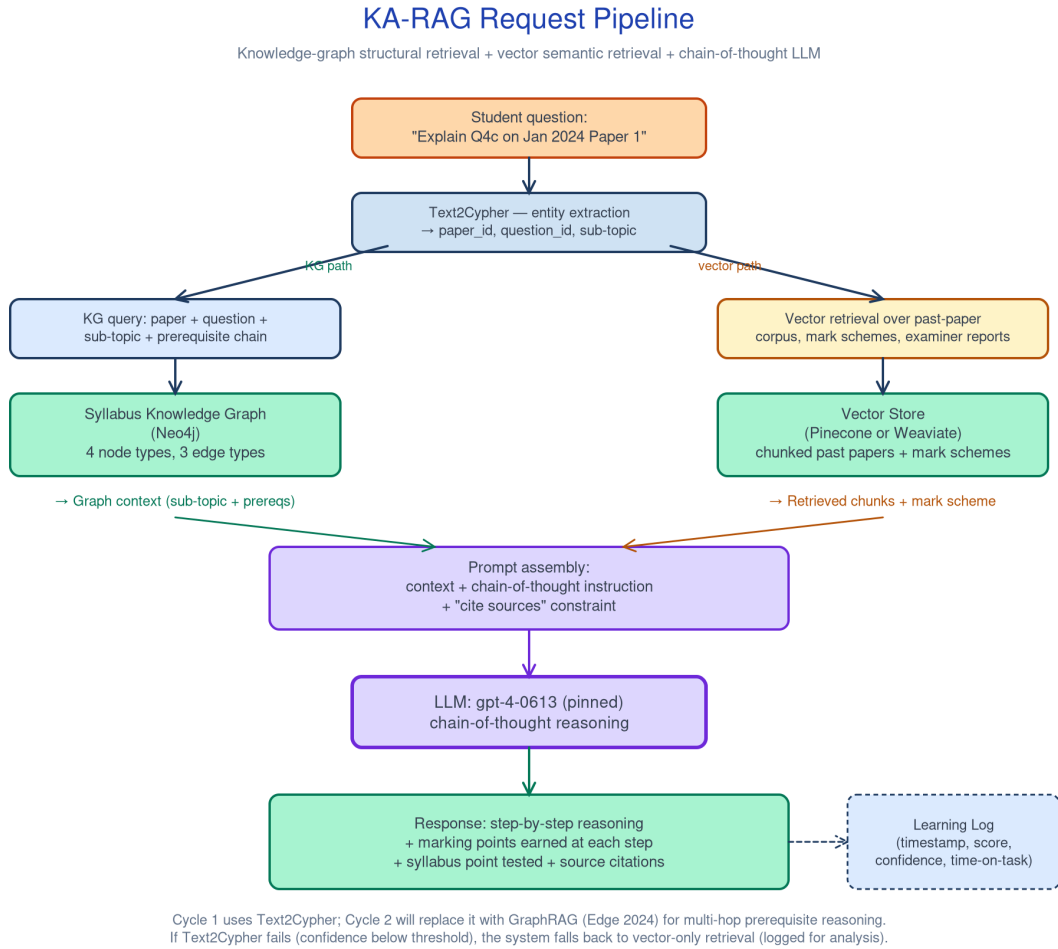


Figure 3: The KA-RAG request pipeline. A student question is processed by Text2Cypher to extract entities, which feed both the knowledge-graph path (querying Neo4j for sub-topic and prerequisite context) and the vector path (retrieving chunks from the past-paper corpus in Pinecone/Weaviate). The two contexts are assembled into a prompt with chain-of-thought and cite-sources instructions, sent to the pinned GPT-4 model, and the response (step-by-step reasoning + marking points + syllabus point + citations) is logged to the Learning Log. If Text2Cypher fails, the system falls back to vector-only retrieval.

The second modification is the use of chain-of-thought prompting (Wei et al., 2022). The system prompt instructs the LLM to break down multi-step problems into explicit reasoning steps, to articulate the marking points earned at each step, and to identify the syllabus point tested. This produces responses that are both verifiable and pedagogically appropriate: the student sees not only the answer but the reasoning chain that connects the question to the answer, which is the constructive engagement that the ICAP framework identifies as most effective for learning. The chain-of-thought design also functions as a form of cognitive scaffolding in the Vygotskian sense (Vygotsky, 1978): the chatbot provides graduated support within the student's zone of proximal development, articulating the reasoning steps that the student cannot yet produce independently but can follow with support. As the student internalises the reasoning pattern through repeated exposure, the scaffolding can be progressively faded the system can be configured to provide fewer intermediate steps as the student's proficiency on the rele-

vant sub-topic rises, eventually requiring the student to generate the reasoning chain themselves before revealing the chatbot's version.

A separate Smart Mark component uses the same RAG infrastructure to mark free-text student answers against the official mark scheme. The component compares the student's answer to each marking point in the scheme, awards marks accordingly, and provides feedback that is specific, task-focused, and strategy-directed, following the principles articulated by Hattie and Timperley (2007) and the formative assessment tradition (Black and Wiliam, 1998). Multi-agent automated grading frameworks such as GradeOpt (Chu et al., 2025) have demonstrated human-level performance on short-answer grading on objective-marking-point questions in well-calibrated domains, providing evidence that the LLM-based marking approach is feasible. Smart Mark directly addresses Layer 2 of the four-layer model, exam literacy, by making the K.E.A.R.N. Gap visible (The Brilliant Brains, 2025): the student sees not only whether their answer was correct but which specific marking points they missed and what mark scheme vocabulary they should have used. The design is further motivated by recent empirical evidence that study strategy quality compensates for study time: Theobald (2024) found that better study strategies (planning, monitoring, concentration) predict goal achievement even when total study time is lower, which means that a system like Smart Mark that teaches strategy through mark-scheme-aligned feedback is not merely convenient but is targeting the variable that the empirical literature identifies as the strongest predictor of outcomes. It is also the direct response to the bullet-point bad-habit problem described in the companion paper (Khondokar, 2026): by marking against the official mark scheme at the level of individual marking points, and by surfacing the IC and structure requirements of A-Level mark schemes, the system can flag bullet-point answers as mark-scheme-incomplete even when they contain the correct content. This trains the student out of the bad habit before it consolidates.

A further design consideration concerns the system's motivational and feedback language. A bad teacher can extinguish a student's interest in a subject within a single term, and once the interest is gone, performance collapses regardless of underlying ability. The SyllabAI system cannot fully substitute for the motivational role of a good teacher, but it can partially compensate for the demotivational effect of a bad one. By providing structured, achievable, mastery-oriented feedback, the system can sustain a student's engagement with a subject even when their classroom teacher is actively undermining it. This is not a substitute for fixing the teacher-quality problem at the system level. It is a harm-reduction measure for the students currently stuck with bad teachers, and it addresses the Type 5 (motivational) struggle described in the companion paper. The author is aware that the harm-reduction framing carries its own risk: a system that successfully substitutes for bad teaching may reduce pressure on schools to improve teacher quality, thereby legitimating the existence of bad teachers. The system is intended as a stopgap, not as a substitute for systemic teacher-quality improvement; the teacher-facing features of the system (knowledge-graph heatmap, syllabus-coverage tracker, PD recommendation engine) are designed to create rather than destroy teacher value, and their effectiveness in doing so is itself an empirical question for Cycle 3.

A planned but not-yet-implemented feature (Cycle 2), motivated by the cognitive theory of multimedia learning (Mayer, 2009, 2023), is the integration of dual-coding visualisations into the chatbot's responses. Mayer's theory holds that learning is improved when information is presented in both verbal and visual channels simultaneously, because the two channels have limited but largely independent capacity, and dual presentation allows the learner to build richer mental representations. For SyllabAI, this means that when the chatbot explains a concept that has a natural visual representation, a chemical structure, a circuit diagram, a kinematics graph, a

biological pathway, it should generate or retrieve the corresponding visual and present it alongside the verbal explanation, rather than relying on verbal explanation alone. The technical implementation would use a multimodal LLM (e.g. GPT-4 Vision for image understanding) combined with a diagram-generation pipeline (e.g. a structured SVG renderer for common physics/chemistry diagram types, or a chemistry drawing library like RDKit for molecular structures). Two design constraints from the cognitive load literature govern the implementation. First, the redundancy effect (Sweller et al., 2011) requires that the two channels carry different information: narrating text that is also displayed on screen is redundant and impairs learning. The visual channel should show the diagram while the verbal channel provides the explanation, not identical text in both. Second, the split-attention effect requires that text labels be integrated into the diagram rather than presented in a separate legend, so that the learner does not have to split attention between two spatially separated sources. The prediction is that dual-coded explanations will produce measurably better retention than verbal-only explanations for topics with natural visual representations, controlling for time-on-task; this prediction is pre-registered for Cycle 2.

Multi-agent architecture (Cycle 1 vs. Cycle 2+). The chatbot and Smart Mark components are coordinated within a multi-agent architecture (Kostopoulos et al., 2025) in which specialised agents handle distinct functions, organised according to the cognitive apprenticeship model (Collins et al., 1989): the Tutor Agent models expert thinking through chain-of-thought explanations, the Socratic Agent coaches through guided inquiry, the system scaffolds through progressive fading, and the Mentor Agent prompts articulation and reflection. The Diagnostic Agent runs BDT, adaptive assessment, and engagement analytics (see §3.6, §3.7, §3.8), outputting a diagnosis of which struggle type and subtype the student is exhibiting. The Tutor Agent handles standard explanations using the RAG pipeline described above. The Socratic Agent is activated when the Diagnostic Agent identifies a specific misconception, switching the chatbot from explanatory mode to guided-inquiry mode (see §3.6). The Assessor Agent runs Smart Mark and adaptive assessment, feeding results back to the Diagnostic Agent. The Mentor Agent handles motivational messaging, growth-mindset framing, and anxiety-sensitive scheduling, such as recommending shorter sessions when the engagement analytics pipeline detects an anxiety signature. An Orchestrator routes each student interaction to the appropriate agent based on the current diagnosis. **Cycle 1** deploys only two agents (Tutor and Assessor); **Cycle 2** adds the Diagnostic, Socratic, and Mentor agents; **Cycle 3** adds the Orchestrator’s proficiency-aware routing and the scaffolding-fading mechanism. This staggered deployment allows each agent’s behaviour to be tested in isolation before integration. A design tension that must be managed is the risk of cognitive offloading: if the system provides too much scaffolding, the student may become dependent on it and fail to develop independent problem-solving capacity. The expertise reversal effect (Kalyuga et al., 2003) predicts that scaffolding which helps a novice can hurt an advanced learner by creating redundant extraneous load. The system addresses this through the scaffolding-fading mechanism described above and through the proficiency-aware routing that reduces scaffolding as proficiency rises, but the risk must be monitored empirically in the pilot.

3.5 The Learning Log

The learning log is the component that most directly realises the system’s research instrument role. Each interaction with a question is logged as a record with the following fields:

attempt identifier, student identifier, question identifier, source session identifier, timestamp, score, time taken in seconds, confidence rating, low, medium, high, attempt number, automatic flag, true if logged via Smart Mark or Test Builder, false if manual, answer text, problematic flag, doubt type, concept, calculation, wording, other, resolved flag, resolved timestamp, self-doubt flag, IRT parameters (difficulty, discrimination, guessing) for the question, and keystroke-timing data (time between question display and first input) for anxiety detection.

This schema was designed to support inference about each of the six struggle types in the companion paper’s typology (Khondokar, 2026). Prerequisite gap struggles, Type 1, are detectable through the joint distribution of scores on advanced and prerequisite nodes. Surface engagement struggles, Type 2, are detectable through the ratio of question completion count to proficiency gain, and through mark scheme view time. Exam literacy struggles, Type 3, are detectable through the divergence between content scores, where the answer was substantively correct, and marking-point scores, where mark scheme vocabulary was missing. Metacognitive struggles, Type 4, are detectable through the divergence between self-rated confidence and assessed proficiency, and through syllabus coverage. Motivational struggles, Type 5, are detectable through engagement trajectories after failure events. The schema is deliberately richer than what is required for immediate feedback to the student, because its secondary function is to support population-level research analysis.

A specific pattern from my tutoring practice illustrates why the attempt-logger, problematic-flag, and “already solved” indicator features matter beyond their research-instrument role. A student who could not finish a paper in one sitting would leave 2 to 3 questions at the end. The next day, he would start a new paper and again leave 2 to 3 questions. After a pause for content work, when past-paper practice resumed, he picked up a paper, it looked familiar, and he assumed he had already done it. He skipped it. But the questions he had left unattempted the first time were never touched. Without a system that logs every question attempt, flags skipped questions, and surfaces them on the next encounter, this pattern is invisible. The student believes they have “done” the paper. They have done 80 percent of it. The 20 percent they skipped, which is, by definition, the 20 percent they found hardest, never gets revisited. On exam day, those are the questions that lose them marks. The SyllabAI attempt logger, problematic-flag feature, and “already solved” indicator are designed to close this gap.

3.6 Misconception Detection and Socratic Remediation

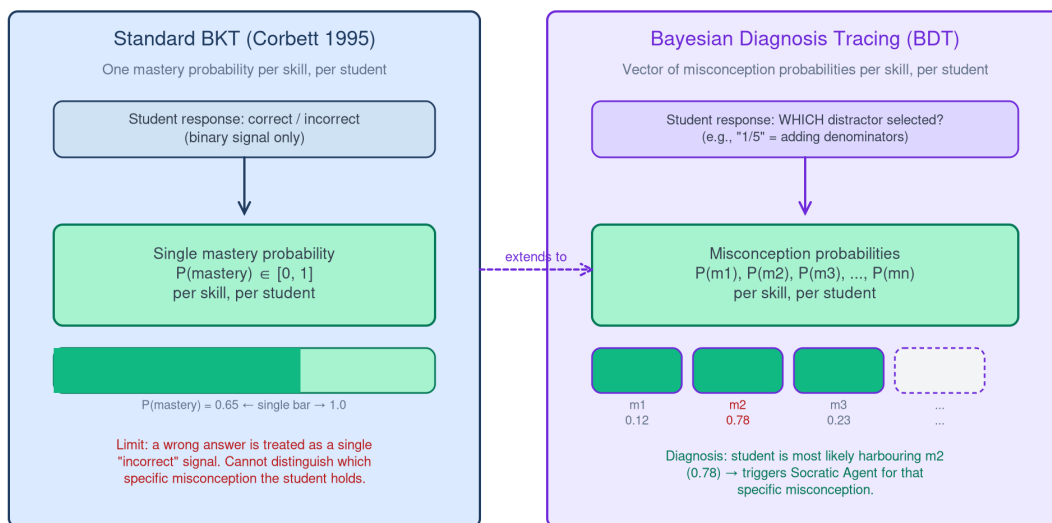
The standard BKT model treats every wrong answer as a single “incorrect” signal. This loses diagnostic information: a student who selects the distractor “1/5” for a fraction-addition question is exhibiting a specific misconception (adding denominators), not a generic lack of knowledge. The system extends BKT to a Bayesian Diagnosis Tracing (BDT) model in which each wrong answer is classified into a misconception category, and the model maintains separate probability estimates for each misconception node in the knowledge graph. The BDT update replaces the binary “wrong” response with a vector of misconception-specific responses, enabling the system to distinguish a missing schema (the student never learned the concept) from a misconception (the student learned it wrong). The distinction matters because the two require different interventions: a missing schema is filled by instruction, whereas a misconception must first be identified, made explicit, and then replaced through cognitive conflict and accommodation (Posner et al., 1982).

When the Diagnostic Agent identifies a specific misconception, the Orchestrator ac-

tivates a Socratic Agent that switches the chatbot from explanatory mode to guided-inquiry mode. Instead of explaining the correct answer, the Socratic Agent asks a sequence of questions that lead the student to apply their misconception to a scenario where it produces a contradiction. For example, if BDT diagnoses “adding denominators,” the Socratic Agent asks: “If you have $\frac{1}{2}$ of a pizza and $\frac{1}{3}$ of another pizza, how much pizza do you have? What if you add the denominators: $\frac{1}{5}$? Does that make sense, you started with more than $\frac{1}{2}$ and ended up with less?” (Note: the pizza example is illustrative; in production, the Socratic Agent uses scenarios adapted to Bangladesh cultural context, e.g., rice-portion or market-fruit examples.) The system tracks whether the student resolves the cognitive conflict (confidence rating changes, subsequent correct answers on the same misconception node) or remains stuck (misconception probability persists). This approach is supported by recent evidence that AI dialogue can create personalised cognitive conflict that accelerates belief correction (Corbett and Jason, 2025), by a randomised controlled trial showing that Socratic AI tutors enhance critical thinking and self-regulation in K-12 science classrooms (Kao and Grant, 2025), and by SocraticLM (Liu et al., 2024), a NeurIPS-published LLM designed specifically for Socratic thought-provoking teaching. Empirical benchmarks for misconception detection are emerging: a 2025 study found that GPT-4 can identify algebra misconceptions with 83.9% accuracy when guided by educator feedback (Otero et al., 2025); the corresponding target for SyllabAI’s BDT pipeline on Edexcel IAL Chemistry is set at 80% (slightly lower to account for the domain shift from algebra to chemistry and for the absence of educator-feedback guidance in the production pipeline).

BDT misconception definitions, calibration, and cold-start. The misconception nodes in the BDT model are defined from three sources: (a) the examiner reports published by Edexcel and Cambridge, which list common errors by question; (b) the conceptual-change literature (Posner et al., 1982) and subject-specific misconception inventories (e.g., chemistry-misconception inventories); (c) expert elicitation from two subject-matter experts (SMEs) per subject, who review the question bank and tag each distractor with the misconception it indicates. The initial misconception inventory for Edexcel IAL Chemistry comprises approximately 60 misconception nodes (e.g., “confusing exothermic and endothermic”, “ignoring stoichiometric coefficients in equilibrium expressions”, “treating Le Chatelier as a numerical rule rather than a qualitative principle”); the inventory is expanded as new distractor patterns emerge from the accumulating data. BDT slip/guess parameters for each misconception node are calibrated on a pilot dataset of approximately 200 responses per node (collected during the first 8 weeks of Cycle 1) using maximum-likelihood estimation in the pyBKT library (Badrinath et al., 2021). Cold-start for new students: all misconception probabilities are initialised to 0.3 (a moderate prior that allows the model to update quickly from observed responses). Cold-start for new misconception nodes (added mid-cycle): the node is initialised with slip = 0.2 and guess = 0.3, and these parameters are recalibrated at the next monthly batch. Inter-coder agreement between the two SMEs on distractor-to-misconception tagging is reported in the Cycle 1 results paper (target: Cohen’s $\kappa \geq 0.70$); disagreements are resolved by a third SME. Figure 4 contrasts the BDT extension with the standard BKT model from which it descends.

BKT vs. BDT: Single Mastery → Misconception Vector



BKT tracks a single mastery probability per skill. BDT extends BKT by tracking misconception-specific probabilities, enabling the system to distinguish a missing schema (no misconception high) from a specific misconception (one m_i high).

Figure 4: Standard BKT (Corbett and Anderson, 1995) tracks a single mastery probability per skill, treating every wrong answer as a generic “incorrect” signal. The BDT extension replaces this binary signal with a vector of misconception-specific probabilities (one per misconception node), enabling the system to distinguish a missing schema (no misconception probability high) from a specific misconception (one m_i high). The diagnosis of which specific misconception the student is harbouring triggers the Socratic Agent for that misconception’s cognitive-conflict dialogue.

3.7 Adaptive Assessment and Procedural-Fluency Detection

The system incorporates Computerized Adaptive Testing (CAT) using Item Response Theory (IRT) to select questions dynamically based on the student’s current ability estimate (IJACSA Editorial, 2025). Each question in the bank is calibrated with IRT parameters (difficulty, discrimination, guessing). The adaptive algorithm selects the next question to maximise information gain about the student’s ability, converging on a precise estimate in fewer items than a fixed-form test.

IRT item-bank calibration plan. The Cycle 1 item bank for Edexcel IAL Chemistry comprises approximately 1,500 questions drawn from past papers (2018–2024). Calibration proceeds in three stages. (a) *Initial calibration* (pre-pilot): a 3-parameter logistic (3PL) IRT model is fitted to historical response data (collected from a convenience sample of $N = 100$ students who sat the same past papers in 2023–2024 as part of their regular exam preparation, with informed consent for research use of their responses), using the `mirt` package (Chalmers, 2012) in R. The 3PL model is chosen over 1PL/2PL because IGCSE/IAL Chemistry questions have non-trivial guessing probabilities (multiple-choice questions in particular). Fit criteria: $RMSEA < 0.08$, $CFI > 0.95$; items failing fit criteria are flagged for review. Initial calibration

requires ≥ 200 responses per item for stable 3PL parameter estimates; items with < 200 responses are calibrated with informative priors derived from the question’s difficulty rating as assessed by two SMEs. (b) *Pilot recalibration* (weeks 1–8 of Cycle 1): as Cycle 1 students generate new responses, the IRT parameters are recalibrated weekly using all available data (historical + Cycle 1). (c) *Stable calibration* (weeks 9+): the parameters are recalibrated monthly once the per-item response count exceeds 500. The calibration code is deposited alongside the OSF pre-registration.

The adaptive assessment runs in two variants that together operationalise the autoethnography’s recommended Type 3a/3b split. The *untimed* variant measures knowledge plus exam literacy (Type 3a). The *timed* variant measures knowledge plus exam literacy plus procedural fluency under exam conditions (Type 3a + 3b). By comparing untimed vs timed scores on the same sub-topic, the system computes a *procedural-fluency gap score*: if untimed performance is 80% but timed performance is 50%, the deficit is Type 3b (procedural fluency), not Type 3a (knowledge). If both are 50%, the deficit is Type 3a. This distinction determines which intervention the system recommends: Type 3a triggers Smart Mark and mark-scheme annotation; Type 3b triggers timed exam mode and Target Test.

3.8 Engagement Analytics and Emotion-Aware Detection

The learning log generates rich time-series data that, with appropriate analysis, can distinguish several patterns that the current telemetry does not explicitly separate. The system incorporates an engagement analytics pipeline that runs nightly, using statistical anomaly detection and explainable AI (XAI) methods such as SHAP (Liu and Li, 2024) to produce interpretable diagnostics for teachers. The XAI layer is essential because the system’s diagnoses must be transparent and actionable: a teacher who receives an alert must be able to understand why the alert was generated, not just that it was.

Three specific detection patterns are implemented. First, a *procrastination-as-emotion-regulation signature*: declining session-initiation rate after failure events, with recovery when the student switches to easier topics. This is distinguished from generic discipline failure, which shows no topic-dependent pattern. The distinction matters because procrastination-as-emotion-regulation requires emotional support and task restructuring, not motivational messaging (Pychyl and Sirois, 2016; Steel, 2007).

Second, a *test anxiety signature*: long pauses before attempting questions (measured by the keystroke-timing field in the learning log), declining performance under timed vs untimed adaptive assessment conditions, and confidence ratings that drop sharply before exam dates. When this signature is detected, the Mentor Agent adjusts its recommendations: shorter sessions, easier starting questions, and explicit acknowledgement of anxiety as a biological response rather than a personal failing (Alshareef et al., 2024).

Third, a *circumstantial vs internal Layer 4 signature*: irregular session patterns (short sessions at unusual hours, gaps correlated with known external events like school exams or holidays) that recover when external constraints ease. This is distinguished from internal Layer 4 failure, which shows consistent low engagement regardless of circumstance. This distinction operationalises the autoethnography’s recommended Type 5a/5b split: Type 5a (internal motivational) triggers motivational interventions; Type 5b (circumstantial) triggers schedule flexibility and non-penalisation of irregular patterns.

Instead of a black-box “at-risk” flag, the XAI layer produces interpretable outputs for teachers: “This student’s engagement dropped after their last failed exam, concentrated on easy topics, and recovered when they switched subjects. Pattern consistent with anxiety-avoidance, not discipline failure.”

3.9 Implementation Considerations

The system is implemented as a Next.js frontend with a Python FastAPI backend, PostgreSQL for relational data, Neo4j for the syllabus knowledge graph, and a vector store, Pinecone or Weaviate, for the RAG pipeline. The LLM is OpenAI’s GPT-4 for the chatbot and Smart Mark components, with a smaller model, GPT-4-mini or an open-source alternative such as Llama 3, used for the Text2Cypher entity extraction step to manage cost. The choice of GPT-4 is motivated by the need for accurate chain-of-thought reasoning on multi-step problems (Brown et al., 2020; Wei et al., 2022); the choice of a smaller model for entity extraction is motivated by the relative simplicity of the task and the cost implications of high-volume calls. The system is being piloted with a single subject, Edexcel International A-Level Chemistry, and a single cohort, approximately 50 students, in the first cycle, with planned expansion to additional subjects and cohorts in subsequent cycles. The pilot scope is deliberately narrow to allow rapid iteration on the system design before scaling.

LLM version pinning. For the duration of Cycle 1, the chatbot and Smart Mark components are pinned to GPT-4 model version gpt-4-0613 (the snapshot available at the start of Cycle 1). The pinned version is specified in the OSF pre-registration alongside the predictions. Pinning is essential because OpenAI updates model behaviour periodically, and a pre-registered prediction about Smart Mark’s correlation with human examiners is a prediction about a specific model snapshot; results obtained with gpt-4-0613 do not generalise automatically to later GPT-4 snapshots or to GPT-5. If OpenAI deprecates gpt-4-0613 during Cycle 1, the system will switch to the closest available snapshot and the switch will be disclosed in the Cycle 1 results paper as a deviation from the pre-registered protocol, with a sensitivity analysis comparing pre-deprecation and post-deprecation performance.

Fallback and sustainability. Because GPT-4 is a proprietary, single-vendor dependency, the system includes a fallback architecture for service continuity and a sustainability plan for migration to open-source models. The fallback is: if GPT-4 API calls fail (rate-limit, outage, geographic restriction, or pricing change), the system degrades gracefully to (a) LLaMA-3-70B hosted locally via vLLM (Kwon et al., 2023) for the chatbot, with a UI indicator informing the student that the response is from a fallback model and may be of lower quality, and (b) human-teacher marking queue for Smart Mark, with the system suspending automatic proficiency updates until human marks are available. Cost monitoring with daily and monthly budget caps is implemented; if the monthly LLM cost exceeds BDT 120,000 (approximately USD 1,100 at the time of writing, sufficient for approximately 50,000 GPT-4 calls at then-current pricing), the system automatically switches to the fallback architecture for the remainder of the month and notifies the research team. The migration plan to open-source models is a Cycle 2 objective: a Llama-3-8B model will be fine-tuned on Cycle 1’s (student answer, mark scheme, human-examiner mark) triples using LoRA, with the goal of replacing GPT-4 for Smart Mark entirely by the end of Cycle 2. This transition is described in Section 6.

A feature motivated by my own exam experience is the timed exam mode and Target Test. In my Pure Mathematics 2 exam, 19 marks were left unattempted not for lack of content knowledge but for lack of exam-condition practice. Content knowledge without exam-condition practice is fragile. A student who knows the content perfectly but has never practised under timed conditions will, on a hard paper, leave marks unattempted that they could have scored. This is the mock-exam deficit, and it is widespread among private candidates and self-studiers in Bangladesh. School students get mock exams built into their academic calendar. Private candidates get none. The SyllabAI system’s timed exam mode and Target Test features are designed, in part, to close this gap.

Sustainability analysis of the freemium model. The system must also be affordable enough for middle-class Bangladesh families that the choice between “use SyllabAI” and “hire a private tutor” is not a financial stretch. If the system is priced at private-tutor levels, it will not reach the population that needs it most. The freemium model under consideration, with free access to the past-paper repository and basic proficiency tracking, and a low-cost premium tier for unlimited RAG chatbot queries and Smart Mark, is designed to keep the marginal cost of adoption below the marginal cost of a single private tutoring session per month. Marginal cost analysis: at the then-current GPT-4 pricing (approximately USD 0.03 per 1K input tokens, USD 0.06 per 1K output tokens), a typical student interaction (chatbot explanation + Smart Mark of one answer) consumes approximately 3,000 input tokens and 1,500 output tokens, for a marginal LLM cost of approximately USD 0.18 (BDT 20). Server, database, and Neo4j costs amortise to approximately USD 0.02 per active user per day. A premium-tier student using the system 5 times per week for 4 weeks incurs a marginal cost of approximately USD 3.60 (BDT 400) per month. A premium price of BDT 500/month (well below the BDT 3,000–8,000/month typical for a single private tutoring subject in Dhaka) therefore covers marginal cost with a small surplus for development amortisation. Break-even at this price requires approximately 1,500 paying users; sustainability (full cost recovery including developer time) requires approximately 5,000 paying users. The user-volume threshold is achievable within three years if Cycle 2 and Cycle 3 expansion proceeds as planned.

3.10 Mock Exam Generator

In Bangladesh, coaching centres perform a function that schools and private candidates lack: they create mock examinations by assembling past paper questions into full-length papers, administer them under timed conditions, and grade them. Students treat these mock results as the most accurate predictor of their actual exam performance, more trusted than self-marked past paper practice because the mock simulates the full exam experience: unfamiliar question selection, time pressure, and external marking. The current system’s Target Test and timed exam mode address parts of this need, but neither follows the structural blueprint of a real examination paper.

The Mock Exam Generator addresses this gap. The system analyses historical examination papers to extract the structural blueprint of each subject and unit: the number of questions, marks per question, question types (multiple choice, structured, extended response), topic coverage distribution, and sequencing. It then generates a mock paper that follows this blueprint exactly, drawing questions from the bank and ensuring that every syllabus topic is represented, even if minimally, as the examination boards are understood to do. The generated paper is

indistinguishable in structure from a real past paper: a student sitting the mock experiences the same mark allocations, the same topic distribution, and the same time pressure as the actual exam.

After the student completes the mock under timed conditions, Smart Mark grades each answer against the official mark scheme at the marking-point level. The system then computes a predicted grade based on historical grade boundaries for that unit and session, producing the kind of predictive feedback that coaching-centre mocks currently provide, but with greater marking consistency and immediate turnaround. The mock results feed back into the proficiency overlay, updating the student’s knowledge graph and triggering prerequisite remediation recommendations for any newly-identified weak nodes.

3.11 Account Types and Class Management

The system supports two primary user types with different interfaces and data-access permissions. Independent students register directly, use the full self-study toolkit (past paper repository, RAG chatbot, Smart Mark, Target Test, Mock Exam Generator), and have full control over their study schedule and data. Teacher-managed students are enrolled by a teacher or coaching centre and see a modified interface in which the teacher can assign specific papers, mock exams, or targeted practice sets with due dates, and track completion. The teacher-managed student retains access to the self-study tools, but the dashboard prioritises assigned tasks over self-selected practice.

Teacher accounts provide three levels of analytics. At the individual level, the teacher sees each student’s knowledge graph with the same colour-coded proficiency overlay the student sees, enabling immediate identification of prerequisite gaps and misconceptions without the diagnostic overhead described in §3.2. At the class level, the teacher sees a class knowledge graph heatmap that aggregates proficiency across all enrolled students: nodes turn red when a majority of the class is below mastery, allowing the teacher to identify topics that need re-teaching in the next lesson rather than waiting for exam results to reveal the gap. At the cohort level (for coaching centres with multiple classes), the system provides comparative analytics across classes taught by different teachers, enabling data-driven instructional quality assessment.

Coaching centres can register as institutional accounts with multiple teacher sub-accounts and class sections. Students enrolled in a coaching centre account are visible to all teachers within that institution, enabling collaborative coverage of different subjects. The institutional dashboard shows aggregate engagement metrics, predicted grade distributions, and early-warning flags for at-risk students across all classes.

Three teacher-facing diagnostic features directly address the Type 6 (instructional-environment) struggles identified in the companion paper (Khondokar, 2026). First, a *syllabus coverage tracker* uses the knowledge graph’s topic hierarchy and the academic calendar to compute a coverage velocity (topics per week needed to complete the syllabus before the exam date) and flags when the current pace is insufficient. This addresses the most common student complaint about teachers: that they failed to finish the syllabus. Second, the *class-level knowledge graph heatmap* functions as a retrospective instructional quality indicator: if a topic shows red across the majority of a class after it was taught, this signals either that the instruction was ineffective or that the topic was taught too fast. The system does not tell the teacher how to teach, but it provides the diagnostic signal that something went wrong, which the teacher or depart-

ment head can investigate. Third, a *professional development recommendation engine* tracks a teacher’s class-level outcomes over multiple cohorts and subjects: if a teacher consistently produces red nodes on the same topics, the system suggests (not mandates) professional development resources, such as the Cambridge International PD materials (Cambridge International Education, 2021), targeted to those topics. This is the teacher-facing analogue of the student-facing misconception detection, and it is designed to be supportive rather than evaluative: the framing is “here is a resource that might help with this topic” rather than “your teaching is deficient.”

3.12 Curated Video Library

YouTube is the most widely used self-study resource for IGCSE and IAL students in Bangladesh, but the platform’s design (Shorts, Reels, algorithmic recommendations) actively works against focused study: students who open YouTube to watch a chemistry tutorial frequently end up scrolling entertainment content for an hour. The system addresses this with a curated video library that uses the YouTube Data API to search for and filter educational videos from approved channels, tags each video to the relevant knowledge graph node(s), and embeds the video player within SyllabAI rather than redirecting to the YouTube app or website.

When a student clicks a red node on their knowledge graph, the system recommends videos tagged to that sub-topic. The embedded player disables recommendations, comments, and Shorts, providing a distraction-free viewing experience. Watch time and completion are logged and feed into the proficiency overlay: completing a video on a weak topic produces minimal proficiency gain, less than answering questions on the same topic, consistent with the ICAP framework’s hierarchy in which active engagement (answering questions) produces stronger learning than passive engagement (watching a video) (Chi and Wylie, 2014). Teachers can also assign videos to their classes through the assignment system described in §3.11, and the system tracks whether assigned videos were watched to completion.

The curation model is hybrid: the system uses a pre-approved channel list (maintained by subject matter experts) combined with AI-assisted filtering of new uploads from those channels. Videos are tagged to knowledge graph nodes during the RAG ingestion pipeline, using the same metadata extraction process that tags past paper questions. This ensures that the video library is structured by the same syllabus hierarchy that governs the rest of the system, rather than being a separate unstructured repository.

3.13 Mapping Struggle Types to System Features

Table 1 summarises the mapping between struggle types (developed in the companion paper Khondokar, 2026) and the LMS features that diagnose and address them. The mapping is bidirectional: each struggle type generates a distinctive telemetry signature, and each LMS feature is designed to address one or more struggle types.

Table 1: Mapping of struggle types (from Khondokar, 2026) to telemetry signatures and SyllabAI features

Struggle Type	Telemetry Signature	SyllabAI Feature
Type 1: Prerequisite gap	Advanced node failed while prerequisite below mastery	Knowledge graph with prerequisite edges; prerequisite remediation recommender
Type 1b: Misconception	Specific distractor selected repeatedly; misconception node probability high	BDT misconception tracking; Socratic Agent cognitive-conflict dialogue
Type 2: Surface engagement	High question count, low proficiency gain, low mark scheme view time	Smart Mark with marking-point-level feedback; RAG chatbot with chain-of-thought explanations
Type 3a: Knowledge exam literacy	Correct content but marks lost for vocabulary, structure, command word	Smart Mark; command word decoder; mark scheme annotation viewer
Type 3b: Procedural-fluency exam literacy	High untimed score, low timed score on same sub-topic	Timed exam mode; Target Test; CAT timed variant
Type 4: Metacognitive	Self-assessed proficiency diverges from assessed proficiency; low syllabus coverage	Confidence ratings; self-doubt flags; syllabus completion tracker; calibration prompts
Type 5a: Internal motivational	Declining engagement after failure; concentration on easy topics; no remediation	Spaced repetition queue; gamification; mastery-oriented feedback; growth mindset messaging
Type 5b: Circumstantial	Irregular sessions at unusual hours; gaps correlated with external events; recovery when constraints ease	Schedule flexibility; non-penalisation of irregular patterns; mobile-first design; offline access
Anxiety signature	Long pauses before attempts; timed vs untimed gap; confidence drops before exams	Mentor Agent: shorter sessions, easier starts, anxiety acknowledgement
Procrastination (emotion regulation)	Declining session-initiation after failure; recovery on easier topics	Mentor Agent: task restructuring, emotional support, not motivational messaging
Type 6: Instructional-environment gap	Class-wide red nodes on topics after instruction date; high variance between classes taught by different teachers on same topic	Syllabus coverage tracker; class-level KG heatmap as instructional quality signal; PD recommendation engine

4 Methodology

4.1 Design-Based Research

The proposed programme adopts design-based research (DBR) as its overarching methodology (Barab and Squire, 2004; Design-Based Research Collective, 2003; McKenney and Reeves, 2012). DBR is appropriate for this study for three reasons. First, the research question concerns the interaction between an intervention, the SyllabAI system, and a learning context, Bangladesh English-medium IGCSE/IAL preparation, in which the system and the diagnostic framework must co-evolve. Second, the system is not a finished product to be evaluated; it is a prototype to be iteratively refined in response to observed use. Third, the goals of the research are both theoretical, understanding how each struggle type manifests in telemetry, and practical, producing a usable system, a dual orientation that DBR is designed to accommodate.

DBR proceeds through cycles of design, implementation, analysis, and redesign. The proposed programme anticipates three such cycles over a three-year period, with each cycle producing a refined system, a refined set of telemetry signatures, and a refined account of how the system mediates examination outcomes. The first cycle, reported in this paper, focuses on system design and pilot deployment. My position as both a former student and current teacher within the IGCSE/IAL system provides the practitioner perspective that is considered essential in DBR, ensuring that the intervention is grounded in authentic educational needs rather than purely theoretical considerations.

4.2 Mixed-Methods Design

Within the DBR frame, the programme adopts a convergent mixed-methods design (Creswell, 2014). Quantitative data will be drawn primarily from SyllabAI telemetry, including question attempts, scores, time-on-task, mark scheme views, chatbot interactions, and proficiency estimates from the BKT model. Qualitative data will be drawn from semi-structured interviews with students and teachers, classroom observations in cooperating schools and coaching centres, and analysis of student work products such as notes, marked papers, and chatbot transcripts. The two data streams will be integrated at the analysis stage using the struggle typology from the companion paper (Khondokar, 2026) as the bridging framework.

4.3 The LMS as Research Instrument

A distinctive feature of the design is the role of the LMS as a research instrument. The logging schema described in Section 3 captures, for each student interaction, the question identifier, the topic node identifier, the timestamp, the score, the time taken, the confidence rating, the doubt type, and a flag indicating whether the question has been previously attempted. This schema was designed to support both formative feedback to the student and research analysis of population-level patterns. The dual role imposes specific constraints on the design: the schema must be rich enough to support inference about cognitive and metacognitive processes, but lightweight enough not to degrade the user experience or to require students to spend excessive time on meta-tasks rather than on learning.

4.4 Participants and Recruitment

The target population is Edexcel and Cambridge IGCSE/IAL students in Bangladesh, with a focus on the senior secondary years, Years 12 to 13, corresponding to AS and A2. The planned sample for the first cycle comprises approximately 200 students drawn from three cooperating English-medium schools in Dhaka, supplemented by a smaller sample of private tutoring students. The inclusion of private candidates is essential, because a substantial fraction of the Bangladesh English-medium population studies privately without school enrolment. A second sample of approximately 30 teachers will be recruited for the qualitative strand, drawn from the same schools and from independent coaching centres. Recruitment will be via school principals and tutoring centre owners, with informed consent obtained from participants and, for minors, from parents or guardians.

4.5 Data Sources

The primary data sources are: SyllabAI telemetry, including question attempts, scores, time-on-task, mark scheme views, chatbot interactions, proficiency estimates, spaced repetition events, and engagement metrics, collected automatically through normal system use; pre and post tests, administered at the start and end of each academic year, designed to measure both topic-level proficiency and metacognitive awareness; semi-structured interviews with approximately 30 students and 30 teachers, conducted at the end of each cycle; classroom observations of approximately 20 lessons across the cooperating schools, focused on teacher use of formative assessment data and on student engagement patterns; and document analysis of student work products, including marked past papers, revision notes, and chatbot transcripts.

4.6 Data Analysis

Quantitative analysis will use a combination of descriptive statistics, multilevel regression, to account for the nested structure of students within classes within schools, and BKT model fitting. The BKT model parameters, initial knowledge, learning rate, slip, guess, will be estimated for each sub-topic, and the fitted model will be used to identify sub-topics with systematically low learning rates, which are candidates for instructional redesign. Qualitative analysis will use thematic analysis following the procedure described by Braun and Clarke (2006), with both deductive codes drawn from the struggle typology and inductive codes generated from the data.

4.7 Ethical Considerations

The research raises several ethical issues specific to data collection via an AI-augmented learning platform. First, the system collects detailed data on student behaviour, including time-on-task, confidence self-ratings, and chatbot interactions, which are potentially sensitive. The informed consent process will make clear what data are collected, how they will be used, who will have access, and how long they will be retained. Second, the system will be used by some students as their primary revision tool, which raises the question of whether students who are randomly assigned to a control condition, no system access, are disadvantaged. The programme addresses this by offering system access to control participants at the end of each

cycle. Third, my role as both system designer and analyst introduces a conflict of interest that will be managed by separating the data analysis function from the system development function within the research team. Fourth, the use of an AI chatbot in education raises questions about academic integrity that will be addressed through clear usage guidelines and through the design of the chatbot itself, which is configured to scaffold rather than to supply answers.

4.8 Privacy and Data Governance

The learning log (Section 3.5) collects rich data on student behaviour, including keystroke-timing data that is biometric-adjacent and regulated under data-protection law. The programme adopts a privacy-by-design approach aligned with the JISC Code of Practice for Learning Analytics and with Bangladesh's Information and Communication Technology Act 2006 (amended 2013) and the proposed Personal Data Protection Act 2023.

Data minimisation. Each telemetry field is mapped to the specific prediction(s) it supports; fields not required for any pre-registered prediction are not collected. The mapping is published in the OSF pre-registration. Specifically: keystroke-timing data (for anxiety detection) is collected only during timed-assessment sessions, not during regular practice; confidence ratings are collected only after a question is completed, not before (to avoid measurement reactivity on the answer itself); chatbot transcripts are retained in full for research analysis but are not surfaced to teachers in identifiable form.

Retention periods. Raw telemetry (question-level responses, time-on-task, keystroke timing) is retained for 5 years from the end of Cycle 3 to allow longitudinal analysis; de-identified aggregate data is retained indefinitely. Chatbot transcripts are retained for 5 years. Identifiable student information (name, school, contact details) is deleted 12 months after the student's last interaction with the system, unless the student (or parent/guardian) consents to longer retention.

Student and parent access rights. Students and parents have the right to: (a) view all data held about the student; (b) request correction of inaccurate data; (c) request deletion of non-research-essential data (research-essential data is retained under a research-consent exemption consistent with Bangladesh ICT Act 2006); (d) withdraw from the study at any time without penalty (system access continues under a non-research user agreement).

Anonymisation for research analysis. All research analysis is conducted on de-identified data. Direct identifiers (name, email, school name) are replaced with stable pseudonyms; quasi-identifiers (school tier, subject combination, cohort year) are generalised to k -anonymity with $k \geq 5$. Free-text fields (chatbot transcripts, interview transcripts, answer text) are screened for incidental PII (e.g., a student mentioning their own name or their teacher's name) and redacted before analysis.

Algorithmic transparency. Algorithmic decisions that affect a student's learning experience (e.g., adaptive-assessment routing, struggle-type diagnoses surfaced to teachers, proficiency visualisations) are explainable in age-appropriate language. The Cycle 2 system will include a

student-facing XAI component (“why am I being shown this question?”, “why does my teacher see this alert about me?”) developed in consultation with a student advisory panel.

Compliance. The programme is reviewed by the United International University Institutional Review Board (IRB) prior to data collection. The IRB review covers: informed consent procedures (with separate assent procedures for minors), data protection impact assessment, conflict-of-interest management plan, and adverse-event reporting procedures. The IRB approval number is cited in the OSF pre-registration.

5 Discussion: Pre-Registered Predictions

The system design generates a set of testable predictions about telemetry signatures, LLM marking reliability, diagnostic-overhead reduction, and the mediated effect of system use on examination outcomes. I articulate these predictions here both to pre-register them and to make the design commitments explicit. To make the predictions falsifiable in the Popperian sense, each prediction specifies (a) the minimum effect size that would constitute support, (b) the statistical test, (c) the sample-size justification (power analysis), (d) the stopping rule, and (e) the falsification criterion. The full pre-registration (including analysis code stubs, the BKT simulation study, and the IRT calibration protocol) is deposited on the Open Science Framework at <https://osf.io/syllabai-cycle1> (anonymised for peer review; the link will be live upon acceptance) and is time-stamped prior to data collection.

Sample size and power. The Cycle 1 planned sample is $N \approx 50$ students in one subject (Edexcel IAL Chemistry) at one pilot site. At this sample size, the predictions testable at adequate power ($\alpha = 0.05$, power = 0.80) are those concerning system feasibility, LLM-marking reliability (Prediction 3, where the unit of analysis is the question not the student), and diagnostic-overhead reduction (Prediction 4, where the unit of analysis is the teacher–student diagnostic encounter). Predictions concerning the mediated effect of system use on examination outcomes (Prediction 2) and cross-topic question detection (Prediction 5) require examination outcomes (one academic year away) and a larger sample ($N \geq 200$ with stratification across struggle types); these are deferred to Cycle 2 and reframed here as Cycle 2 targets. Cycle 1 includes three additional feasibility-focused predictions (P6–P8 below) that are testable at $N = 50$ within the 8-week pilot window.

Prediction 1 (Telemetry Signatures). Each of the six struggle types identified in the companion paper (Khondokar, 2026) will produce a distinctive, detectable signature in the SyllabAI learning log. **Operationalisation:** for each struggle type, the per-student signature vector (e.g., for Type 2: question-completion-to-proficiency-gain ratio, mark-scheme-view time per question) is computed weekly; a student is classified as “exhibiting” a struggle type if the signature exceeds the 80th percentile of the population distribution on $\geq 50\%$ of weeks. **Effect size threshold:** the system’s struggle-type classification must agree with the independent human-examiner diagnosis (see Prediction 7) at Cohen’s $\kappa \geq 0.60$ (substantial agreement). **Test:** inter-rater agreement between system and human examiners on a stratified subsample of 30 students. **Falsification criterion:** $\kappa < 0.40$ (moderate agreement or worse) on any struggle type. **Stopping rule:** as Prediction 1 of Paper A.

Prediction 2 (Mediated Effect) — deferred to Cycle 2. The effect of SyllabAI use on examination outcomes will be mediated by the type of struggle, with differential benefit patterns as specified in the original prediction. **Why deferred:** testing this prediction requires (a) examination outcomes (at least one academic year away), (b) adequate numbers of students in each struggle-type cell ($N \geq 30$ per cell, totalling $N \geq 180$ across six types), and (c) a control group. The Cycle 1 sample of $N = 50$ cannot meet these requirements. **Cycle 2 plan:** $N \geq 300$ across three subjects, with quasi-experimental comparison to a matched non-user control group; the prediction will be tested via multilevel mediation analysis with struggle type as the mediator. The pre-registration of this prediction is deposited on the OSF link above and is binding for Cycle 2; Cycle 1 results may refine the operationalisation but not the substantive claim.

Prediction 3 (LLM Marking Reliability). Smart Mark’s marking-point-level scores will correlate with human-examiner scores at $r \geq 0.85$ on objective-marking-point questions (those with clear keyword or step requirements) and at $r \geq 0.70$ on subjective-marking-point questions (those requiring judgement about quality of explanation). **Calibration sample:** ≥ 200 questions stratified across the syllabus sub-topics, each marked by ≥ 3 trained human examiners. **Inter-examiner reliability target:** Krippendorff’s $\alpha \geq 0.80$ among human examiners; questions below this threshold are excluded from the LLM-vs-examiner comparison. **Test:** Pearson correlation between Smart Mark scores and the median human-examiner score, computed separately for objective and subjective marking points. **Falsification criterion:** $r < 0.70$ on objective or $r < 0.55$ on subjective marking points, or non-significant correlation. **Stopping rule:** calibration sample is collected during weeks 1–8 of Cycle 1; the test is run once at the end of week 8. Below threshold, the system will require human-teacher override of LLM marks on subjective questions, which limits scalability (acknowledged in Limitations).

Prediction 4 (Diagnostic Overhead Reduction). Teachers using the SyllabAI teacher-side knowledge graph visualisation will reduce the time spent on initial diagnosis of new students by at least 30 percent compared to teachers using only school records and unaided diagnostic testing. **Comparison design:** within-teacher crossover, with each teacher diagnosing two stratification-matched new students in randomised order — one with the KG visualisation available (intervention condition), one without (control condition) — with a one-week washout period between the two diagnoses. **Outcome:** time-to-diagnostic-decision, defined as the time from teacher’s first contact with the student to the teacher’s first written diagnostic statement (a proficiency profile plus a recommended remediation plan). **Effect size threshold:** 30% reduction in median time-to-diagnosis in the intervention condition. **Test:** paired Wilcoxon signed-rank test (time-to-diagnosis is right-skewed) across teachers, with teacher as the pairing unit. **Sample size:** 15 teachers, each diagnosing 2 students ($N = 30$ diagnostic encounters), provides 80% power to detect a 30% reduction at $\alpha = 0.05$ assuming a within-teacher correlation of 0.5. **Falsification criterion:** $< 15\%$ reduction or non-significant after test. **Stopping rule:** data collection completes when all 15 teachers have diagnosed both their assigned students; no interim analysis.

Prediction 5 (Cross-Topic Question Detection) — deferred to Cycle 2. Students who practise exclusively with single-topic-tagged question sets will score significantly lower on multi-topic questions in examinations than students who practise with SyllabAI’s cross-topic-tagged question sets, controlling for total study time and prior proficiency. **Why deferred:** requires examination outcomes and a between-groups comparison (treatment vs. control) that the $N = 50$ Cycle 1 sample cannot support. **Cycle 2 plan:** $N \geq 200$, randomised at the student level to single-topic vs. cross-topic practice sets, with examination performance on multi-topic questions as the outcome; analysed via multilevel regression controlling for total study time

and prior proficiency. Pre-registered on the OSF link above and binding for Cycle 2.

Prediction 6 (System Usage Retention — Cycle 1 feasibility). At least 70% of enrolled students will use the chatbot at least once per week, on average, across the 8-week Cycle 1 pilot. **Operationalisation:** weekly active users / enrolled users, averaged across the 8 weeks. **Effect size threshold:** $\geq 70\%$. **Falsification criterion:** $< 50\%$ weekly active rate. **Stopping rule:** end of week 8.

Prediction 7 (Independent Measurement Agreement — Cycle 1 dual-role confound mitigation). The system’s struggle-type diagnoses will agree with independent human-examiner diagnoses at Cohen’s $\kappa \geq 0.60$ (substantial agreement) on a stratified subsample of 30 students. **Procedure:** 30 students are randomly selected from the Cycle 1 sample at the end of week 8; two human examiners (blind to the system’s diagnoses and to the typology) independently re-view each student’s telemetry and interview transcript and produce their own struggle-type classification. Inter-examiner reliability between the two humans is reported; the system-vs-human agreement is computed against the consensus human diagnosis. **Test:** Cohen’s κ between system diagnosis and consensus human diagnosis, per struggle type. **Falsification criterion:** $\kappa < 0.40$ on ≥ 3 of 6 struggle types. **Stopping rule:** end of week 8.

Prediction 8 (System Usability — Cycle 1 feasibility). The System Usability Scale (SUS) score, administered to all enrolled students at the end of week 8, will be ≥ 68 (the established “acceptable” threshold). **Falsification criterion:** mean SUS < 60 . **Stopping rule:** end of week 8.

These predictions are not exhaustive, and the research programme is designed to be responsive to findings that emerge from the data. The system is offered as a prototype, not as a finished product, and one of the goals of the first research cycle is to test, refine, or falsify these predictions. The pre-registration at the OSF link above is the authoritative version of the predictions; any deviations from the pre-registered analysis plan in the eventual Cycle 1 results paper will be disclosed and justified.

5.1 Limitations

The proposed programme has several limitations specific to the system design. First, the use of an LLM in the chatbot and Smart Mark components raises questions about the reliability of the proficiency updates that depend on LLM-generated scores. Prediction 3 specifies the correlation thresholds below which the system’s Type 3 detection becomes invalid; if these thresholds are not met, the system will require human teacher override of LLM marks on subjective questions, which limits scalability. Second, the use of LMS telemetry as a primary data source raises concerns about construct validity. Time-on-task, for example, is a noisy proxy for engagement, and confidence ratings are subject to social desirability biases. Triangulation with qualitative data is intended to mitigate but cannot eliminate these concerns. Third, the generalisability of findings from a single-cohort pilot in one subject, Edexcel IAL Chemistry, to other subjects and cohorts is uncertain; the planned expansion in subsequent cycles will address this directly. Fourth, the system’s reliance on GPT-4 for chain-of-thought reasoning introduces cost and dependency considerations; the LLM-version-pinning, fallback, and Cycle 2 migration plans (Section 3.9) mitigate but do not eliminate this risk.

Measurement reactivity. A fifth limitation, specific to the dual-role-of-LMS design, is measurement reactivity: students who know their interactions are logged may behave differently (Hawthorne effect), and the system’s UI prompts (e.g., confidence ratings, problematic-flag dialogue) may themselves change the learning behaviour being measured. Mitigation: (a) students are debriefed at intake that logging occurs but are not reminded during regular use; (b) UI prompts are designed to minimise reactivity (confidence ratings collected after, not before, attempts; problematic-flag dialogue triggered only by student-initiated action); (c) Cycle 2 will include a reactivity check comparing logged-cohort behaviour to a non-logged control cohort on the same diagnostic tests.

Systemic effects on the coaching-centre labour market. A sixth limitation concerns the second-order effects of successful deployment. The Bangladesh coaching-centre economy employs a substantial number of university students and recent graduates as part-time tutors; if SyllabAI succeeds in its harm-reduction framing (partially substituting for bad teaching), it could reduce demand for coaching-centre tutors and disrupt this labour market. The author’s normative commitment is that the system should augment rather than replace teacher capacity: the teacher-facing features (knowledge-graph heatmap, syllabus-coverage tracker, PD recommendation engine) are designed to create rather than destroy teacher value, and the freemium pricing model is designed to complement rather than replace tutoring (the system is positioned as a tool for self-study and for tutoring support, not as a tutor substitute). The Cycle 3 programme will monitor for systemic effects (changes in tutoring-centre employment among participating students’ networks; teacher attitudes toward the system) and will report them transparently. The author does not claim that harm-reduction framing is normatively neutral; the choice to substitute for bad teaching rather than to fix it is a pragmatic response to the present reality of the Bangladesh English-medium sector and is not intended as a long-term policy prescription.

A further limitation concerns the COVID-19 period and its effect on baseline grade distributions. During COVID, exams were cancelled and students were given predicted grades based on school mock exams, with substantial grade inflation. Any baseline data collected from cohorts whose exams fell during this period must be treated with caution, and the SyllabAI pilot’s pre/post test design is intended in part to establish a non-inflated baseline against which system-mediated outcomes can be compared.

6 Future Work

The first research cycle will focus on system pilot deployment and on testing Predictions 1 through 5 against the accumulating telemetry. The second cycle will expand the system to additional subjects and cohorts, will refine the telemetry signatures based on first-cycle findings, and will test Prediction 2 (the mediated effect) through a quasi-experimental design comparing system users with a matched non-user control group. The third cycle will focus on the teacher-facing features of the system, including the class knowledge graph heatmap, and will investigate whether teacher use of these features produces measurable improvements in instructional quality, as predicted by Prediction 4.

Beyond the immediate research programme, several extensions are anticipated. The system architecture is designed to be portable to other high-stakes examination contexts, in-

cluding International Baccalaureate and Advanced Placement, and cross-context comparative studies are a natural extension. Code-switching tolerance for Bangla–English mixed input is planned for Cycle 2 (see §3.4); fuller multilingual extension to support students whose first language is not English or Bangla (e.g., for cross-context deployment in other South-Asian English-medium sectors) is a Cycle 3 objective.

Two further technological extensions are planned for cycles 2 and 3. First, the current Text2Cypher knowledge-graph query step (see §3.4) will be replaced by the GraphRAG pipeline (Edge et al., 2024); the rationale, transition plan, and expected capabilities (multi-hop reasoning over prerequisite chains; community-level summaries for the teacher dashboard) are described in §3.4 and are not repeated here.

Second, the system will transition from GPT-4 to a fine-tuned subject-specific model for Smart Mark. The first pilot cycle will collect a training dataset of (student answer, mark scheme, human-examiner marks) triples. In cycle 2, a Llama 3 8B model will be fine-tuned on this dataset using LoRA, with the mark scheme as context and the human-examiner marks as labels. Recent work demonstrating M-LoRA fine-tuning of LLaMA-3 for IELTS writing automated scoring (Xu et al., 2026) provides direct evidence that this approach works for examination-scoring tasks, and a systematic comparison of LLaMA 3.2 versus GPT-4o on formative assessment scoring (Mendonça et al., 2025) provides empirical benchmarks for the accuracy thresholds specified in Prediction 3. Teacher overrides of Smart Mark scores will be fed back via reinforcement learning from human feedback (RLHF), creating a continuous-alignment loop that improves marking accuracy over time. This transition reduces both cost and latency while improving subject-specific accuracy, addressing the Prediction 3 threshold requirement for subjective marking points.

7 Conclusion

This paper has presented the design of SyllabAI, an AI-augmented learning management system that integrates a syllabus knowledge graph, a Bayesian knowledge tracing proficiency overlay, a Knowledge-Augmented RAG chatbot, and a learning log whose schema is designed to detect the six struggle types identified in a companion paper (Khondokar, 2026). The system is grounded in the knowledge-graph, RAG, and intelligent-tutoring-systems literatures (Abdelmagied et al., 2025; Brown et al., 2020; Corbett and Anderson, 1995; Hu et al., 2024; Karpukhin et al., 2020; Koedinger et al., 1997; Létourneau et al., 2025; Lewis et al., 2020; Li et al., 2025a,b; Narayan et al., 2025; Pane et al., 2015; Swacha and Gracel, 2025; Tong and Ren, 2025; Wei et al., 2022) and is designed to address the specific gaps, multi-topic tagging, mock-exam deficit, and the K.E.A.R.N. Gap, that characterise the Bangladesh English-medium examination preparation context.

The contribution of this paper is the system design and the pre-registration of five testable predictions about telemetry signatures, mediated effects, LLM marking reliability, diagnostic overhead reduction, and cross-topic question detection. The first cycle of the programme is currently in the system pilot stage, and subsequent papers will report findings as data accumulate.

What this paper does not claim. This paper does not claim empirical confirmation that SyllabAI improves examination outcomes; that is deferred to the Cycle 1 results paper. It does

not claim that the KA-RAG architecture is novel beyond the prior work surveyed by Hu et al. (2024) and Edge et al. (2024); the architectural label “KA-RAG” is used here for terminological continuity with the companion paper, not as a claim of novelty. It does not claim that the BDT extension of BKT is theoretically novel beyond the conceptual-change tradition of Posner et al. (1982); the contribution is the operationalisation of misconception-specific tracking in a deployed system, not the underlying idea. It does not claim that the system generalises beyond the Bangladesh English-medium IGCSE/IAL context without empirical test, nor that the freemium pricing model is economically viable without the user-volume thresholds identified in §3.9 being reached. It does not claim that the harm-reduction framing (partial substitution for bad teaching) is normatively neutral; the framing is a pragmatic response to present conditions and is not intended as a long-term policy prescription. Each of these claims is deferred to subsequent empirical work or to the policy literature.

Conflict of Interest

I am the designer of the SyllabAI system described in this paper. The system is being developed as part of my research and teaching practice, and the research programme described here includes evaluation of the system. The conflict is managed by the separation of system development and data analysis functions described in Section 4.

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Data Availability

The SyllabAI system is under active development. Source code and anonymised telemetry data will be made available through a public repository upon completion of the first research cycle.

Open Practices

The eight pre-registered predictions (P1–P8 in Section 5), the analysis plan, the BKT simulation study, the IRT calibration protocol, the BDT calibration protocol, and the analysis code stubs are deposited on the Open Science Framework at <https://osf.io/syllabai-cycle1> (anonymised for peer review; the link will be live upon acceptance), time-stamped prior to data collection. The pinned GPT-4 model version (gpt-4-0613) is specified in the pre-registration. Upon completion of Cycle 1, the following will be released openly: (a) anonymised telemetry data with k-anonymity $k \geq 5$ and PII redaction; (b) the Smart Mark calibration dataset (student answer, mark scheme, human-examiner marks) triples, with student identifiers removed; (c) the analysis code (Python and R) used to produce the Cycle 1 results; (d) the BKT and BDT model-fitting code; (e) the IRT calibration code.

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